

Accessible Slides in LaTeX

https://github.com/rhstanton/accessible_LaTeX

Version 1.4

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Why this matters (and what we'll cover)

- From [April 2026](#), course materials must meet [WCAG](#) 2.1 Level AA accessibility standards
 - Applies even to password-protected course sites (e.g., [bCourses](#)).
 - Campus guidance issued Jan 8, 2026.
- **Bad news:** Standard $\LaTeX$ (e.g., [Beamer](#)) does not automatically generate accessible PDFs.
- **Good news:** The $\LaTeX$ Project team has developed solutions.
- **Roadmap (4 slides worth of ideas):**
 1. What changes for $\LaTeX$ to make accessible PDFs
 2. Setup requirements (kernel/engine) and Overleaf guidance
 3. Examples: figures, tables, math
 4. Validation workflow + common “false positive” warnings

What is this project?

- A practical template set for accessible $\text{\LaTeX}$ using the [\LaTeX Tagging Project](#)
 - **Two templates:** Articles (**article** class) and slides (**ltx-talk**)
 - **Shows:** What is shared and what differs
 - **Contains:** Math, text, figures, and tables
 - **Validation status:** Slides score 100% with **Ally** and pass common validators (**veraPDF**, **PAC**, **Acrobat**) in typical workflows (tested March 17, 2026).
- Repo: https://github.com/rhstanton/accessible_LaTeX

Converting to accessible $\text{\LaTeX}$: the big picture

- **Your existing $\text{\LaTeX}$ skills transfer:** mainly adding a few consistent accessibility features.
- **Most changes are local:** metadata once; then per-figure and per-table tweaks.
- **Applies to both slides & articles:**
 - Document metadata + PDF tagging
 - Alt text on images
 - Table header row tagging
 - Sufficient color contrast
 - Compile with Lua $\text{\LaTeX}$
- **Key fork:**
 - Articles: **keep your existing class** and add tagging features
 - Slides: switch to `ltx-talk` (frame syntax similar to Beamer), then recreate styling

Version requirements: what actually matters

- **Engine:** must compile with [Lua \$\LaTeX\$](#) (not pdf $\LaTeX$ or Xe $\LaTeX$).
- **LaTeX kernel/format date:** must be [2025-11-01 or later](#).
- **Important:** a “TeX Live year” alone is not a reliable guarantee.
 - You need a sufficiently new $\LaTeX$ kernel *on your install*, or use Overleaf Labs.
- **Quick check (in any document preamble):**

```
\show\fmtversion
```

Overleaf (works well when configured correctly)

- `ltx-talk` requires a very recent $\text{\LaTeX}$ kernel (format date [2025-11-01+](#)).
- On Overleaf, this typically means using the [Labs](#) program (not default projects).
- **Overleaf setup (2 steps):**
 1. Join Overleaf Labs:
 - <https://www.overleaf.com/labs/participate>
 - Enable [Rolling TeX Live releases](#)
 2. Configure the project:
 - TeX Live: [Rolling TeXLive \(labs\)](#)
 - Compiler: [Lua \$\text{\LaTeX}\$](#)
- Overleaf guide: <https://docs.overleaf.com/writing-and-editing/creating-accessible-pdfs>

Core requirement 1: DocumentMetadata (minimal template)

- Required for both slides and articles.
- Must appear **before** `\documentclass`.
- **Minimal pattern (don't overthink it):**

```
\DocumentMetadata{  
  lang=en-US,  
  tagging=on  
}
```

- In practice, you'll usually keep the full template settings (next slide) for PDF/A + PDF/UA and math behavior.

Core requirement 1 (recommended): full metadata block

- This template's \DocumentMetadata block:
 - targets PDF/A-2u + PDF/UA-1
 - enables tagging
 - keeps math accessible and reduces some validator complaints
- **Recommendation:** copy/paste the full block and **leave it unchanged.**

```
\DocumentMetadata{
  pdfstandard=a-2u,          % PDF/A-2u format
  pdfstandard=ua-1,          % Add PDF/UA-1 conformance target
  pdfversion=1.7,            % Explicit PDF version
  lang=en-US,                % Language for screen readers
  tagging=on,                % Enable PDF tagging

  tagging-setup={
    math/alt/use,             % Keep math accessibility enabled
    math/mathml/AF=false,    % Do not embed MathML as PDF Associated Files
    math/tex/AF=false,       % Do not embed TeX source as Associated Files
    math/mathml/sources=     % Clear MathML source attachment list
  }
}
```


Core requirements 2–4: images, tables, and color contrast

- **Required for both slides and articles.**
- **Images:** every `\includegraphics` needs `alt={...}`.
- **Tables:** mark header rows using `\tagpdfsetup{table/header-rows={...}}`.
- **Colors:** WCAG 2.1 AA requires [4.5:1](#) contrast ratio for normal text.
- **Accessible color defaults:**

```
\colorlet{AccessibleRed}{red!80!black}  
\colorlet{AccessibleGreen}{green!40!black}
```

Core requirement 5: compile with Lua $\text{\LaTeX}$

- **Required for both slides and articles.**
- Compile with **Lua $\text{\LaTeX}$** (not pdf $\text{\LaTeX}$ or Xe $\text{\LaTeX}$).
- **Why Lua $\text{\LaTeX}$?**
 1. Automatic math accessibility support (no manual MathML files in normal workflows)
 2. Full tagging support
 3. Good Unicode/OpenType font support
- **How to switch:**
 - Command line: `lualatex filename.tex`
 - Editors: select LuaLaTeX as the compiler

Slides content: mostly the same as Beamer

- Slides live in frame environments (similar to Beamer).
- So your **content** often transfers with minimal changes.
- Here's some gratuitous *math* for accessibility checkers.

- **Typical frame skeleton:**

```
\begin{frame}{Title}  
  \begin{itemize}  
    \item First point  
  \end{itemize}  
\end{frame}
```

- **Note (optional):** fonts in this template keep digits consistent between text and math.

Figures: alt text

- Figures are included as usual (e.g., `\includegraphics`), but **add alt text**.

- Good (minimal):** `\includegraphics[alt={A capybara}]{capybara.jpg}`

- Better (more informative):**

```
\includegraphics[alt={Photo of a capybara's head facing left}]{capybara.jpg}
```

- Decorative images: use empty alt text: `alt={}`.



Tables

- Tables use tabular as usual, but you must specify **header rows**.
- Use {1} for one header row, {1,2} for two header rows, etc.
- **Note:** PAC may still warn about header associations in some cases; see [Validator-specific notes](#).

```
\tagpdfsetup{table/header-rows={1,2,3}}  
\begin{tabular}{cccrcccr}  
\toprule  
← 3 header rows  
\midrule  
← data rows  
\bottomrule  
\end{tabular}
```

Payment date	Caplet expiry date	DF_{pay}	Forward rate	Days to expiry	Days in accrual period	T_{expiry}	Δ	Caplet
2004/12/01	—	0.99550	0.01790	0	91	0.00000	0.25278	—
2005/03/01	2004/11/29	0.99008	0.02188	89	90	0.24384	0.25000	1,178.77
2005/06/01	2005/02/25	0.98401	0.02413	177	92	0.48493	0.25556	4,844.73
2005/09/01	2005/05/27	0.97733	0.02675	268	92	0.73425	0.25556	10,016.71

One checklist (applies to everything)

- **Do these for any accessible $\text{\LaTeX}$ PDF:**
 1. Add `\DocumentMetadata{...}` (before `\documentclass`)
 2. Compile with [Lua \$\text{\LaTeX}\$](#)
 3. Add `alt={...}` to every `\includegraphics`
 4. Add `table/header-rows={...}` before every table
 5. Use sufficient color contrast (4.5:1 for normal text)
- **Useful link:** contrast checker <https://webaim.org/resources/contrastchecker/>

The one big fork: articles vs. slides

- **Articles:** keep your existing class (e.g., `article`) and add the checklist items.
- **Slides:** switch to `ltx-talk`.
 - Your `\begin{frame}` content often stays the same.
 - Remove Beamer themes/templates; recreate styling with standard $\text{\LaTeX}$ packages (as in this template).
- **If you're converting Beamer slides:**

<code>\documentclass{ltx-talk}</code> (instead of beamer)

Getting started (new vs. converting): keep it simple

- **New documents:**

- copy `accessible_slides.tex` or `accessible_article.tex`
- replace content; keep the metadata/preamble structure

- **Converting existing documents:**

- add metadata + Lua^AT_EX + alt text + table header rows
- for slides, switch to `ltx-talk` and redo styling once

- Repo: https://github.com/rhstanton/accessible_LaTeX

Common pitfalls (high-impact mistakes)

- **Confusing PDF tagging with PDF bookmarks**
 - `tagging=on` creates the structure tree for **screen readers** (required).
 - PDF bookmarks (navigation pane) are separate and optional.
- **Forgetting alt text**
 - Every `\includegraphics` needs `alt={...}` (or `alt={}` if decorative).
- **Not tagging table header rows**
 - Add `\tagpdfsetup{table/header-rows={...}}` before each table.
- **Low contrast colors**
 - Avoid light colors (e.g., plain yellow/cyan).
 - Use darker variants for red/green (defined in this template).
- **Wrong engine or old kernel**
 - Use Lua $\text{\LaTeX}$ and ensure format date 2025-11-01 or later.

Validation workflow: use multiple checkers

- No single checker catches everything.
- Accessibility usability checks and strict PDF conformance checks are not identical.
- Practical workflow: run 2+ tools and compare findings.
- **Common tools:**
 - Ally (bCourses)
 - veraPDF (PDF/A conformance): <https://verapdf.org/>
 - PAC (PDF accessibility): <https://pac.pdf-accessibility.org/>
 - Adobe Acrobat (useful but has known quirks)

Optional cleanup for strict validators: strip_af.py

- The metadata settings alone can be enough for a 100% score in Ally.
- Some strict tools (e.g., [veraPDF](#)) may still complain about /AF or embedded-file structures.
- To remove these, run:

```
python strip_af.py build/accessable_slides.pdf
```

- Install prerequisite, [pikepdf](#): `python -m pip install pikepdf`
- Then validate again with veraPDF.

Validator-specific notes (common false positives)

- **Acrobat: “Lbl and LBody – Failed” (validator bug)**
 - Often appears around figure captions, section numbers, table captions.
 - This is an Acrobat bug in common setups; typically safe to ignore if source tagging is correct.
 - Reference: <https://tex.stackexchange.com/a/709655/2388>
- **PAC table warning (known behavior)**
 - PAC can report: Table header cell has no associated subcells even when table/header-rows is correct.
 - Documented by the Tagging Project:
<https://github.com/latex3/tagging-project/issues/1056>
- **Bottom line:** treat one-tool warnings as investigation prompts, not definitive errors.

TL;DR

- **To make an accessible $\LaTeX$ PDF:**
 1. Add `\DocumentMetadata{...}` (before `\documentclass`)
 2. Compile with [Lua \$\LaTeX\$](#)
 3. Every figure: `\includegraphics[alt={...}]{...}`
 4. Every table: `\tagpdfsetup{table/header-rows={...}}`
 5. Use high-contrast colors (4.5:1) and validate with 2+ tools
- Template repo: https://github.com/rhstanton/accessible_LaTeX
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